

First-Order DE Applications in Computer Engineering

Differential Equations – Assignment

Presented by:

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- Analysis of Motherboard BIOS Battery lifespan using the Exponential Decay Model to determine critical failure thresholds.

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- Modeling the spread of the Code Red Worm across internet servers using the Logistic Growth Equation under limited capacity.

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References

- Citation of real-world data sources (Panasonic Datasheets, CAIDA Report, & Tech Reviews).

First Problem: Growth & Decay

Motherboard BIOS Battery Analysis (CR2032)

The Scenario

- **Context:** Computer motherboards rely on CR2032 batteries to retain BIOS settings when unplugged.
- **Issue:** Battery voltage decays naturally over time due to chemical structure.
- **Goal:** Determine the service life before voltage drops below the critical threshold (2.0V).

Mathematical Model

Equation: Exponential Decay Model

$$V' = -kV$$

Given Data:

- **k :** Decay constant
- **t :** Time (Years)
- **Initial Voltage:**

$$V(0) = 3.0$$

- **After 5 Years:**

$$V(5) = 2.5$$

- **Target (Critical):**

$$V_{critical} = 2.0$$

First Problem: Growth & Decay

Motherboard BIOS Battery Analysis (CR2032)

Solution:

Step 1: General Solution

Solving the separable equation

$$V' = -kV$$

Separate the variables

$$\frac{1}{V} dV = -k dt$$

Integrate both sides:

$$\int \frac{1}{V} dV = \int -k dt$$

$$\ln|V| = -kt + C$$

Exponentiate both sides (base e):

$$|V| = e^C \cdot e^{-kt}$$

Let e^C be the general constant A :

General Equation:

$$V(t) = A \cdot e^{-kt}$$

First Problem: Growth & Decay

Motherboard BIOS Battery Analysis (CR2032)

Solution:

Step 2: Finding Constants

Finding the Initial Constant (A)

Using the information that voltage is 3.0V at $t = 0$:

$$V(0) = A \cdot e^{-k(0)} = 3.0$$

$$A \cdot 1 = 3.0 \Rightarrow A = 3$$

The equation becomes:

$$V(t) = 3 \cdot e^{-kt}$$

Finding the Decay Constant (k)

Using the information that voltage is 2.5V after $t=5$ years:

$$V(5) = 3 \cdot e^{-5k} = 2.5 \Rightarrow e^{-5k} = \frac{5}{6}$$

Take the natural logarithm (ln) of both sides to isolate k:

$$k = -\frac{1}{5} \ln\left(\frac{5}{6}\right) \Rightarrow k \approx \mathbf{0.03646}$$

The equation becomes:

$$V(t) = 3 \cdot e^{-0.03646t}$$

First Problem: Growth & Decay

Motherboard BIOS Battery Analysis (CR2032)

The Result:

Step 3: Critical Threshold Calculation

Finding time (t) when voltage drops to 2.0V:

$$2.0 = 3 \cdot e^{-0.03646t}$$

Take the ln of both sides:

$$t = \frac{\ln(2/3)}{-0.03646} \approx 11.1217 \dots$$

The total service life is approximately
11.12 Years

Second Problem: Population Models

Code Red Worm Spread (Logistic Growth)

The Scenario

- **Event:** In 2001, Code Red worm attacked Microsoft IIS servers.
- **Constraint:** The number of weak servers is finite (Limited Capacity).
- **Goal:** Calculate time to infect 99% of the internet infrastructure.
- **Why 99%:** Mathematically, the Logistic Growth model never reaches 100% in finite time.

Mathematical Model

Equation: Logistic Growth Model

$$P' = rP \left(1 - \frac{P}{K} \right)$$

Given Data:

- **$P(t)$:** Number of infected servers
- **Capacity (K):** 360,000 Servers
- **r :** Infection rate || **t :** Time (Hours)

– Initial:

$$P(0) = 100$$

– Observation:

$$P(12) = 150,000$$

Second Problem: Population Models

Code Red Worm Spread (Logistic Growth)

Solution:

Step 1: Separation of Variables

Solving the separable equation

$$\frac{dP}{dt} = rP \left(\frac{K - P}{K} \right)$$

Separate the variables

$$\frac{dP}{P(K - P)} = \frac{r}{K} dt$$

Step 2: Partial Fraction Decomposition

$$\frac{1}{P(K - P)} = \frac{A}{P} + \frac{B}{K - P}$$

$$1 = A(K - P) + BP$$

A and B:

$$A = B = \frac{1}{K}$$

The equation becomes:

$$\frac{1}{K} \left(\frac{1}{P} + \frac{1}{K - P} \right) dP = \frac{r}{K} dt$$

Second Problem: Population Models

Code Red Worm Spread (Logistic Growth)

Solution:

Step 3: Integration

$$\int \frac{1}{K} \left(\frac{1}{P} + \frac{1}{K-P} \right) dP = \int \frac{r}{K} dt \Rightarrow$$

Canceling $1/K$ from both sides:

$$\int \left(\frac{1}{P} + \frac{1}{K-P} \right) dP = \int r dt$$

(Note: By using chain rule for $\int \frac{1}{K-P}$ we can find $-\ln|K-P|$)

$$\ln \left| \frac{P}{K-P} \right| = rt + C$$

Step 4: General Solution

Simplifying for P:

$$\frac{P}{K-P} = A e^{rt} \Rightarrow P(t) = \frac{A e^{rt} K}{1 + A e^{rt}}$$

Dividing by $A e^{rt}$:

$$P(t) = \frac{K}{(A e^{rt})^{-1} + 1}$$

Finding Constants (A) & (r):

By $P(0) = 100$: $A = 3599^{-1}$

By $P(12) = 150,000$: $r \approx 0.654$

Second Problem: Population Models

Code Red Worm Spread (Logistic Growth)

The Result:

Step 5: Finding Time for 99% Infection

Target: $P(t) = 0.99 K = 356,400$

$$356,400 = \frac{360,000}{1 + 3599 e^{-0.654t}}$$

Simplifying for t :

$$t = -\frac{12.783}{-0.654} \approx 19.55$$

Total time to infect 99% of the internet:
Approximately 19 and a half hours.

Third Problem: Cooling Problems

MacBook Pro Thermal Throttling (Newton's Law of Cooling)

The Scenario

- **Context:** Apple's 2018 MacBook Pro (i9) suffered from severe overheating due to a faulty fan curve (Throttling Scandal).
- **Issue:** The processor instantly hits **100°C** under load. When stopped, fans are too slow to cool it down quickly.
- **Goal:** Calculate the waiting time required for the CPU to cool down to a safe **50°C** to regain Turbo Boost performance.

Mathematical Model

Equation: Newton's Law of Cooling

$$T' = -k(T - T_m)$$

Given Data:

- Initial Temp:

$$T(0) = 100^{\circ}\text{C}$$

- After 1 Min:

$$T(1) = 90^{\circ}\text{C}$$

- Ambient Temp (T_m):

$$T_m = 25^{\circ}\text{C}$$

- **k :** Cooling constant.
- **t :** Time (Minutes)

Third Problem: Cooling Problems

MacBook Pro Thermal Throttling (Newton's Law of Cooling)

Solution:

Step 1: Separation of Variables

Solving the separable equation

$$T' = -k(T - T_m)$$

Separate the variables

$$\frac{1}{T - 25} dT = -k dt$$

Step 2: Integration both sides:

$$\int \frac{1}{T - 25} dT = \int -k dt$$

$$\ln(T - 25) = -kt + C$$

Simplifying for T :

$$T - 25 = e^{-kt+C}$$

$$T = e^C \cdot e^{-kt} + 25$$

e^C will be the general constant A :

General Solution:

$$T(t) = 25 + A \cdot e^{-kt}$$

Third Problem: Cooling Problems

MacBook Pro Thermal Throttling (Newton's Law of Cooling)

Solution:

Step 3: Finding Constants

Finding Constant A:

By $T(0) = 100^\circ\text{C}$

$$100 = 25 + A \cdot e^{-k(0)}$$

$$100 = 25 + A$$

$$\mathbf{A = 75}$$

Finding Constant k:

By $T(1) = 90^\circ\text{C}$

$$90 = 25 + 75 \cdot e^{-k(1)}$$

$$e^{-k} = \frac{13}{15}$$

(ln) of both sides to isolate k :

$$k = -\ln\left(\frac{13}{15}\right)$$

$$\mathbf{k \approx 0.143101}$$

Third Problem: Cooling Problems

MacBook Pro Thermal Throttling (Newton's Law of Cooling)

The Result:

Step 4: Required time t to reach 50°C.

Target: $T(t) = 50$

$$50 = 25 + 75 \cdot e^{-kt}$$

Simplifying for t :

$$t \approx \frac{-1.0986}{-0.1431}$$

$$t \approx 7.67 \text{ minutes}$$

The user needs to wait approximately
7 minutes and 40 seconds to cool down

References

Problem I: BIOS Battery (Datasheet)

- Panasonic Corporation (2021): *Coin Type Lithium Manganese Dioxide Battery (CR2032) Data Sheet*. Industrial Solutions Company.
- Link: [Panasonic CR2032 Datasheet \(PDF\)](#)

Problem II: Code Red Worm (Research)

- CAIDA Analysis (2002): Moore, D., et al. "Code-Red: A case study on the spread and victims of an Internet worm." The Center for Applied Internet Data Analysis.
- Link: [CAIDA Analysis of Code-Red](#)

Problem III: Thermal Throttling (Tech Review)

- Dave2D (2018): *Technical Review: "MacBook Pro 15 (2018) - Beware the Core i9"*.
- Link: [Dave2D Original Review](#)
- The Verge (2018): Warren, T. "Apple apologizes for MacBook Pro throttling bug and releases fix."
- Link: [Apple apologizes for throttling Intel CPUs inside MacBook Pros](#)

Thanks !

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